The Green Hub - Sustainable Community Center
MBA Team Final Project - Group 5 - July 6, 2025
RBC MBA - Project Management - Instructor: Dr. Ehab EL Nezamy

About

The Green Hub is a community center powered entirely by renewable energy, combining construction, engineering, and sustainability. It exemplifies integrating project management with sustainable development, providing environmental, social, and economic benefits to the community.

Renewable energy
Smart technologies
Environmentally Responsible Design

The project aims to inspire greener communities and create a lasting positive impact, while fostering a culture of green living.

The Green Hub serves as a sustainable community center fueled by renewable energy and advanced smart technologies. It combines environmental education, community services, and eco-friendly design to generate lasting social, environmental, and economic benefits.

Team

Deliverable 1 - Scope Statement

Component
Content

Description
Sustainable community center promoting environmental awareness and green living: renewable-energy-powered facility with smart technologies, educational workshops, recycling programs, coworking spaces, community activities.

Requirements
Solar PV system - smart lighting & HVAC controls - rainwater harvesting - recycling center - Building Management System (BMS).

Deliverables
Approved charter - architectural & engineering designs - completed sustainable building - renewable energy system - Smart BMS - community training program - final handover documentation.

Boundaries
IN: site prep, building construction, solar PV, rainwater, smart lighting/HVAC, native-plant landscaping, recycling center, training rooms, grand opening. OUT: other cities, residential, commercial retail, post-handover operation.

Assumptions
Funding approved - permits on time - materials available - skilled contractors available - stable weather.

Constraints
$1.2M budget - 40 weeks - LEED standards - local regulations - site limitations.

Strategy
Waterfall-dominant life cycle: sequential initiation, planning, design, procurement, construction, testing, and handover. The Smart Building Management System (BMS) is delivered incrementally through configuration, integration, testing, and commissioning. Agile is not used because the overall scope, budget, and 40-week timeline are fixed.

Acceptance criteria
<= 40 weeks - budget variance < 5% - LEED certified - smart systems operational - client satisfaction ≥ 90% - zero lost-time accidents.

Cost breakdown
Direct activity costs: Design $60K - Construction $570K - Solar PV $120K - Smart Systems/BMS $80K - MEP $140K - Landscaping $40K - Furniture $70K = $1,080K. Contingency reserve: $78K. Management reserve: $42K. Total approved budget: $1,200K.

Life Cycle Selection & Justification

Waterfall is the governing life cycle for the sequential physical works because the requirements, approvals, construction sequence, acceptance criteria, budget, and completion date are defined in advance.

Incremental delivery is limited to the BMS: core monitoring is configured first, followed by HVAC and lighting integration, renewable-energy data, dashboards, and final commissioning. Each increment adds usable functionality against a predefined scope.

Agile is not selected because the project does not allow an evolving overall scope or an open-ended timeline. Controlled

changes are processed through formal change control.
Deliverable 2 - WBS
Deliverable 3 - OBS
Title / Organizational Role
Seniority Level
Assigned Member / Source
Report to
Sponsor / Account director
Executive Level
Internal company appointment
Executive Board
Project Manager
Top Management
Armia
Sponsor
Planning & Scheduling Lead
Lead Level
Ehab
Project Manager
Scope & Risk Officer
Lead Level
Amira
Project Manager
Finance & Partners Officer
Lead Level
Salwa
Project Manager
QA & Governance Officer
Lead Level
Shehab
Project Manager
QA & Audit Assistant
Junior Level
Michael
QA & Governance officer
Procurement & Supply Chain Lead
Lead Level
Internal company appointment
Project Manager
Sustainability & BMS Engineer
Senior Level
Internal company appointment
Project Manager
HSE Lead officer
Senior Level
Internal company appointment
Project Manager
Design, site, MEP and civil engineers
Senior Level
External consultants / contractor
Project Manager
Junior site & Field Engineers
Junior Level

Contractor
Project Manager
Site Supervisors & Foramen
Supervisory Level
External main contractor
Project Manager
Skilled trades & Equipment operators
Skilled Workforce
External main contractor
Site supervisor
General Construction Workers
General Workforce
External main contractor
Foramen
Equipment & Resource Breakdown
Category
Resource / Equipment
Inside Company
Outside Company
Total / Basis
Management
Sponsor, PM, controls, finance, procurement, QA/QHSE, sustainability/BMS, community liaison
9
0
9 people
Design & engineering
Architects, civil/structural, MEP, solar and landscape engineers
0
8
8 people
Construction supervision
Construction manager, senior engineers, site engineers and supervisors
0
10
10 people
Workforce
Skilled trades, equipment operators and general workers
0
32
32 peak workers
Specialists
BMS integration, testing/commissioning and LEED support
1
5
6 people
Office resources
Laptops, PM software, document control and meeting facilities
10 sets
0
Company-owned
Heavy equipment
Excavator, loader, mobile crane, compactor, concrete pump/mixer and dump trucks
0

8 units
Rented/contractor-owned
Technical equipment
Surveying sets, PV installation tools and MEP/BMS test kits
2 sets
9 sets
11 sets
Safety & temporary works
PPE sets, scaffolding, edge protection and site barriers
0
40 PPE + site sets
Contractor-provided

Resource basis: internal staff govern and control the project; external consultants and contractors execute design, construction, specialist installation, and peak site labor. Counts are peak planning estimates and remain within the approved direct-cost baseline.

Deliverable 4 - Project Budget

PMBOK cost-aggregation case study: activity costs are aggregated first; contingency reserves are then added to establish the cost baseline, followed by management reserve to establish the approved project budget.

Operational Cost Levels

Activity Cost Level: $1,080K in estimated direct costs for labor, equipment, materials, and professional services.

Work Package Level: $38K activity contingency is added above activity costs, producing $1,118K in work-package cost.

Control Account Level: a further $40K work-package contingency is added above work-package cost, producing the $1,158K cost baseline.

Green Hub Budget Matrix
Cost Aggregation Level
Amount
Reserve Ratio
Cumulative
Direct Activity Costs
$1,080K
-
$1,080K
Activity Contingency Reserve
+$38K
3.52% of activity cost
$1,118K
Work Package Cost
$1,118K
-
$1,118K
Work Package Contingency Reserve
+$40K
3.58% of work-package cost
$1,158K
Cost Baseline / Control Accounts
$1,158K
-
$1,158K
Management Reserve
+$42K
3.63% of cost baseline
$1,200K
Total Approved Project Budget

$1,200K
10.00% of approved budget is reserves
$1,200K

Financial Overview

Total Approved Budget: $1,200,000

$1,080,000 (90.0%): direct activity execution, resources, and technical deliverables.

$78,000 (6.5%): total contingency reserves for known risks across activities and work packages.

$42,000 (3.5%): management reserve for unforeseen risk events.

Deliverable 5 - Stakeholder Analysis

Stakeholders were assessed using four clear categories only: H H, H L, L H, and L L. Customer/Investor, Sponsor, and Project Manager are treated as separate stakeholders. Municipality/Local Authorities are H L-not H H-because they have high regulatory power but comparatively low day-to-day interest beyond compliance and permits.

Quadrant Summary

H H - Manage Closely: Customer/Investor - Sponsor - Project Manager - Main Contractor - Design Consultant

H L - Keep Satisfied: Municipality/Local Authorities - Environmental Authority

L H - Keep Informed: Site Engineers & Supervisors - Skilled and General Workers - Local Community - Key Suppliers

L L - Monitor: General Public - Non-critical Vendors

Project Schedule Overview - Details in the Excel Sheet

The table and Gantt chart use the same start/end weeks. FS governs the major phase gates; SS with a stated lag is used only for approved overlaps. No SF relationship is required in this high-level schedule.

Phase / WBS Branch	Start	End	Duration	Predecessor / Relationship	WBS Ref
Project Management (continuous)	1	40	40 weeks	Project start (SS)	Branch 1
Initiation	1	2	2 weeks	Project start	1.1
Planning	3	5	3 weeks	Initiation (FS)	1.2-1.4
Design & Engineering	6	11	6 weeks	Planning (FS)	Branch 7
Procurement	12	15			

4 weeks
Design & Engineering (FS)
Branch 8
Site Preparation
16
19
4 weeks
Procurement (FS)
Branch 2
Building Construction
18
28
11 weeks
Site Preparation (SS + 2 weeks)
Branch 3
Green Systems Installation
24
32
9 weeks
Building Construction (SS + 6 weeks)
Branch 4
Landscaping & Outdoor Areas
29
35
7 weeks
Building Construction (FS)
Branch 5
Community Programs & Launch
33
40
8 weeks
Green Systems (SS + 9 weeks)
Branch 9
Testing
36
38
3 weeks
Landscaping (FS); Green Systems complete
6.1-6.2
Handover
39
40
2 weeks
Testing (FS)
6.3-6.4
The Green Hub 40 Week Gantt Chart
Risk Mitigation Sheet for The Green Hub
#
Risk
Probability
Impact Cost (USD)
Impact Time
Owner

Mitigation / Cost Effect
Solution Cost (USD)
Type
1
Building permit delay
High
$17,600 (1.47%)
4 wks
Design Lead
Submit at 70% design; appoint permitting consultant. Cost is added to prevention/consultancy.
$6,000
MITIGATE
2
Late delivery of imported solar equipment
High
$27,400 (2.28%)
5 wks
Procurement Manager
Order early; dual-source; penalty clause and expediting allowance.
$8,000
TRANSFER
3
Material breakage during handling, storage, or contractor handover
Medium
$22,000 (1.83%)
2 wks
Construction Manager
Inspection at receipt; protected storage; handling method statement; supplier/contractor liability.
$5,000
TRANSFER
4
Contractor underperformance or default
Medium
$70,400 (5.87%)
8 wks
Project Manager
Prequalify; performance bond; retention; milestone payments. Bond/admin premium affects direct cost.
$12,000
MITIGATE
5
Scope creep from unapproved requests
High
$25,400 (2.12%)
3 wks
Project Manager
Formal change control and baseline approval; allowance covers administration and impact analysis.
$3,000
MITIGATE
6
Utility connection / net-metering delay
Medium
$8,800 (0.73%)
4 wks

Sustainability Engineer
Apply early; temporary generator/testing arrangement.
$7,500
MITIGATE
7
Local cement and steel price inflation
High
$35,200 (2.93%)
0 wks
Procurement Manager
Although reporting is in USD, local purchases remain exposed to EGP supplier inflation and USD-equivalent escalation.
Bulk-buy, fixed-price contracts, and secure storage.
$10,000
MITIGATE
8
Site safety incident
Medium
$19,600 (1.63%)
2 wks
QHSE Officer
Insurance; permit-to-work; edge protection; PPE and weekly toolbox talks.
$15,000
MITIGATE
Total Potential Cost Impact
Total Potential Delay
Total Mitigation Cost
Reserve Check
$226,400 (18.87%)
28 weeks (worst case, unmitigated)
$66,500 (5.54%)
$66,500 <= $78,000 contingency reserve